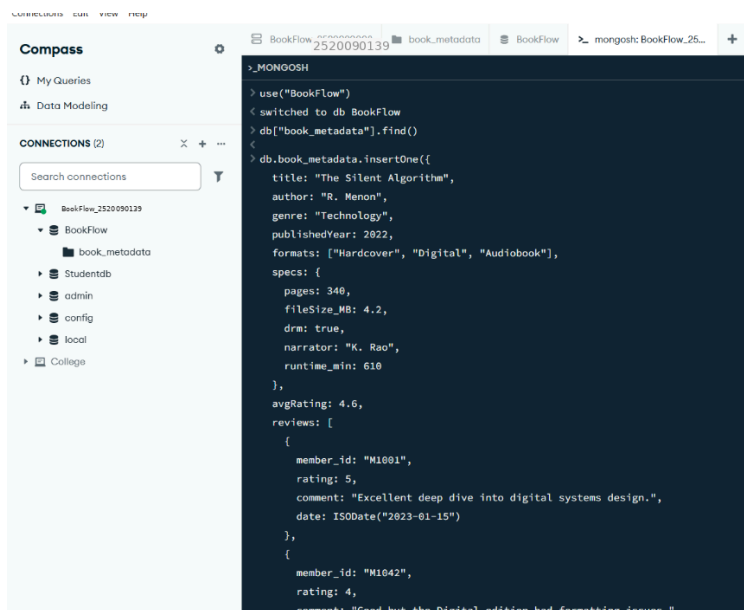


WEEK-3 Practical

2520090139-A.Shashank Reddy

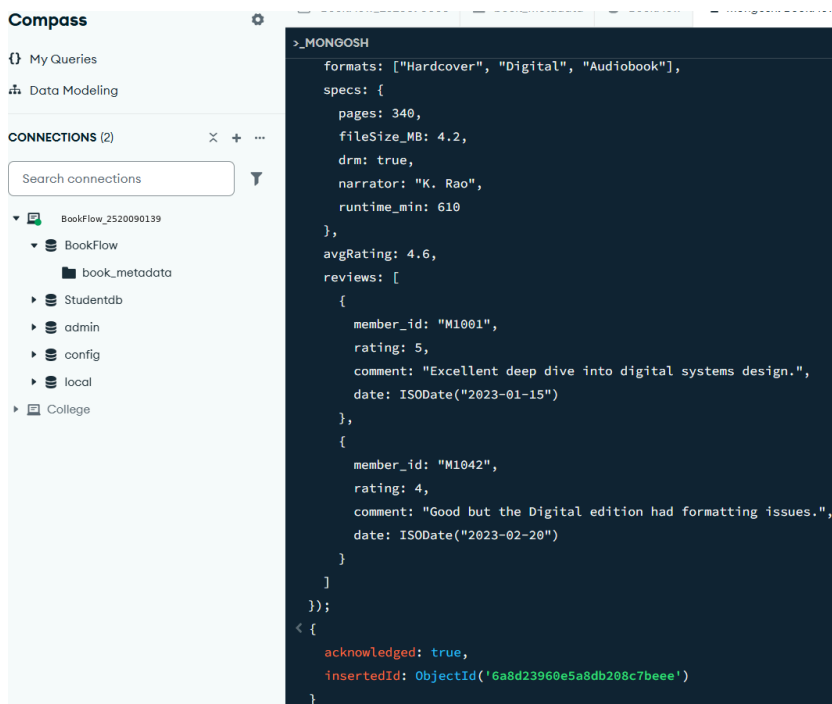
Task-1:

Inserting the First Book Document

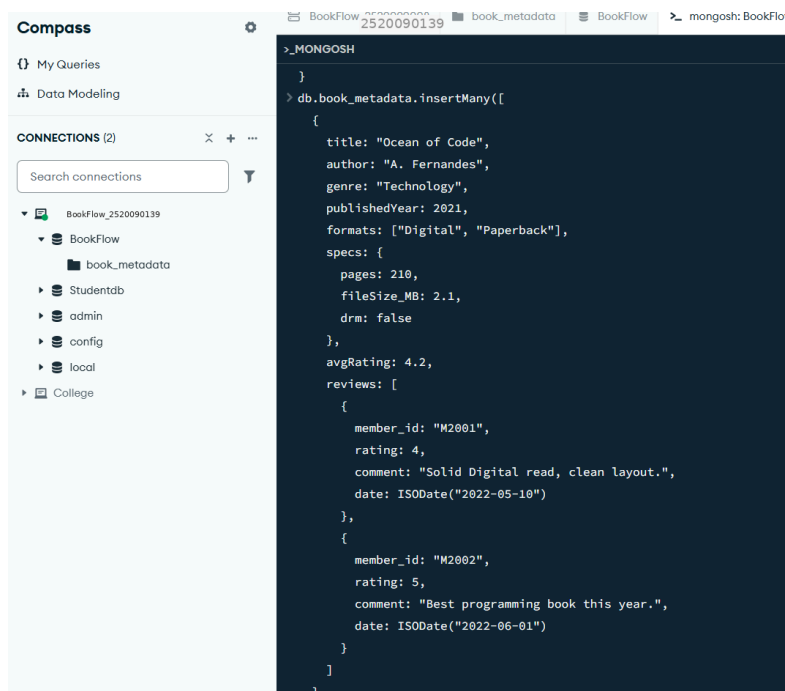


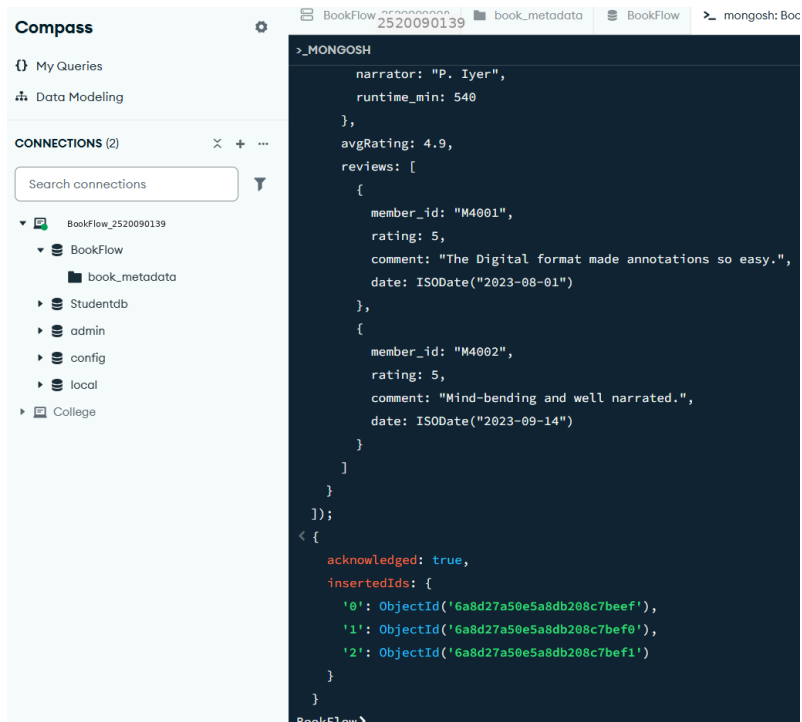
The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists a connection named 'BookFlow_2520090139'. Under this connection, the 'BookFlow' database is expanded, showing a collection named 'book_metadata'. The main panel on the right displays a MongoDB shell session with the following commands and output:

```
> use("BookFlow")
switched to db BookFlow
> db["book_metadata"].find()
[]
> db.book_metadata.insertOne({
  title: "The Silent Algorithm",
  author: "R. Menon",
  genre: "Technology",
  publishedYear: 2022,
  formats: ["Hardcover", "Digital", "Audiobook"],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: "K. Rao",
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: "M1001",
      rating: 5,
      comment: "Excellent deep dive into digital systems design.",
      date: ISODate("2023-01-15")
    },
    {
      member_id: "M1042",
      rating: 4,
      comment: "Good but the Digital edition had formatting issues."
    }
  ]
})
```

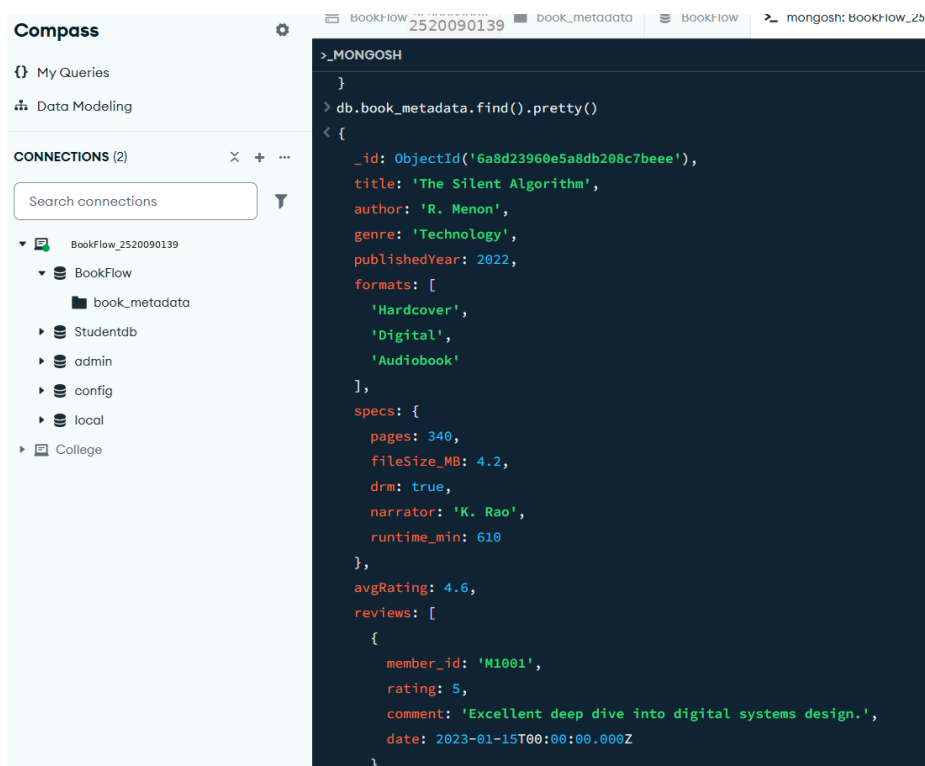


Inserting Multiple Book Documents



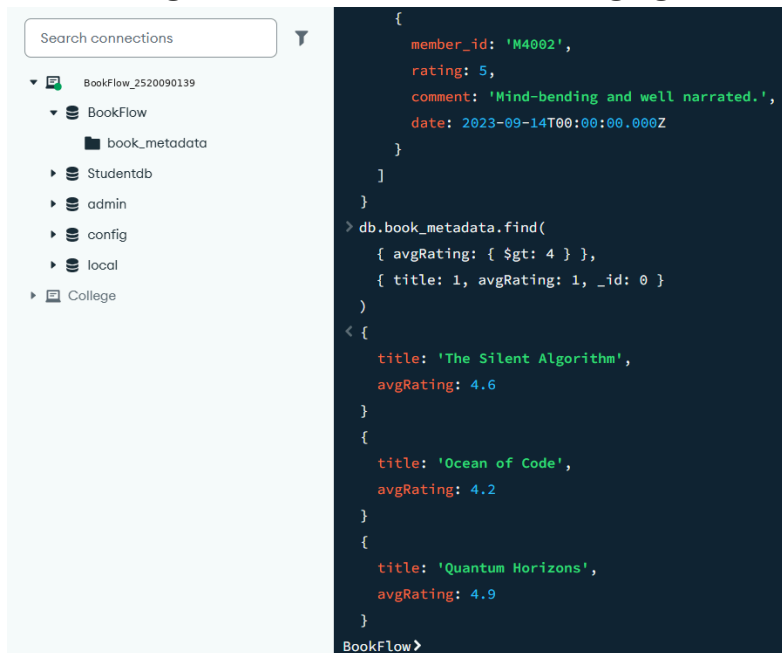


Displaying the Book Metadata Collection



TASK – 2

2.1 Finding Books Rated Above 4 Using \$gt



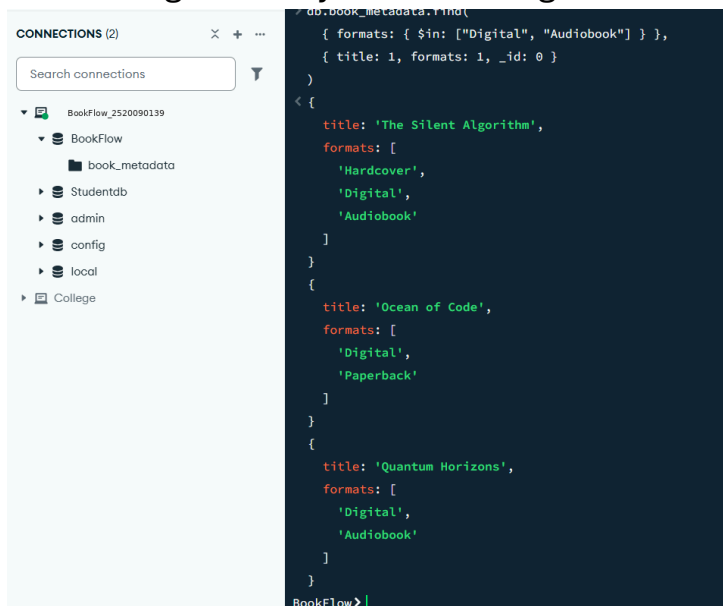
The screenshot shows a MongoDB interface with a left sidebar containing a tree view of connections. The 'BookFlow' connection is selected, and the 'book_metadata' collection is expanded. The main area displays a MongoDB query and its results.

```
Search connections
```

- BookFlow_2520090139
 - BookFlow
 - book_metadata
 - Studentdb
 - admin
 - config
 - local
 - College

```
> db.book_metadata.find(
  { avgRating: { $gt: 4 } },
  { title: 1, avgRating: 1, _id: 0 }
)
< {
  title: 'The Silent Algorithm',
  avgRating: 4.6
}
{
  title: 'Ocean of Code',
  avgRating: 4.2
}
{
  title: 'Quantum Horizons',
  avgRating: 4.9
}
BookFlow>
```

2.2 Finding Books by Format Using \$in



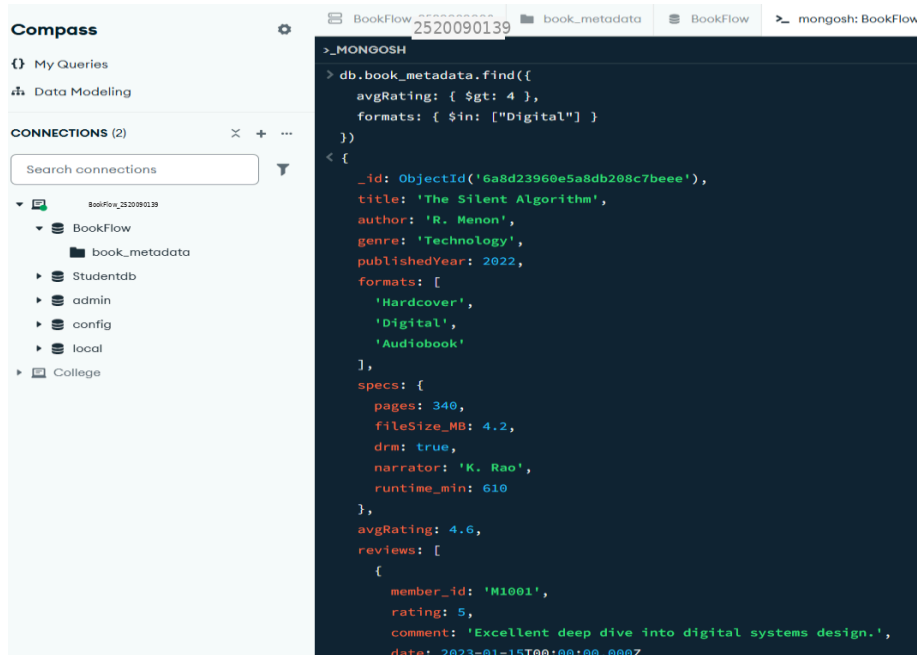
The screenshot shows a MongoDB interface with a left sidebar containing a tree view of connections. The 'BookFlow' connection is selected, and the 'book_metadata' collection is expanded. The main area displays a MongoDB query and its results.

```
CONNECTIONS (2) Search connections
```

- BookFlow_2520090139
 - BookFlow
 - book_metadata
 - Studentdb
 - admin
 - config
 - local
 - College

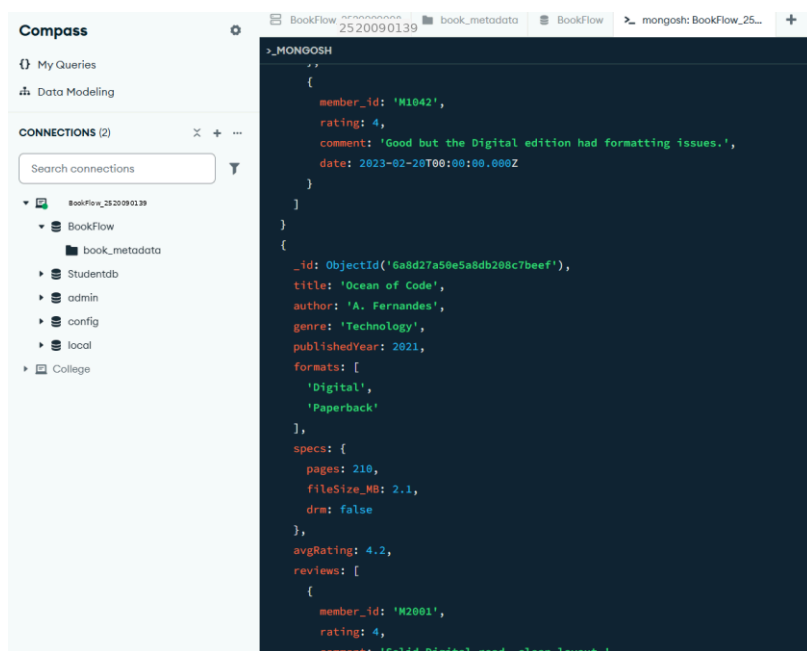
```
> db.book_metadata.find(
  { formats: { $in: ["Digital", "Audiobook"] } },
  { title: 1, formats: 1, _id: 0 }
)
< {
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
{
  title: 'Ocean of Code',
  formats: [
    'Digital',
    'Paperback'
  ]
}
{
  title: 'Quantum Horizons',
  formats: [
    'Digital',
    'Audiobook'
  ]
}
BookFlow>
```

2.3 Combining Rating and Format Conditions



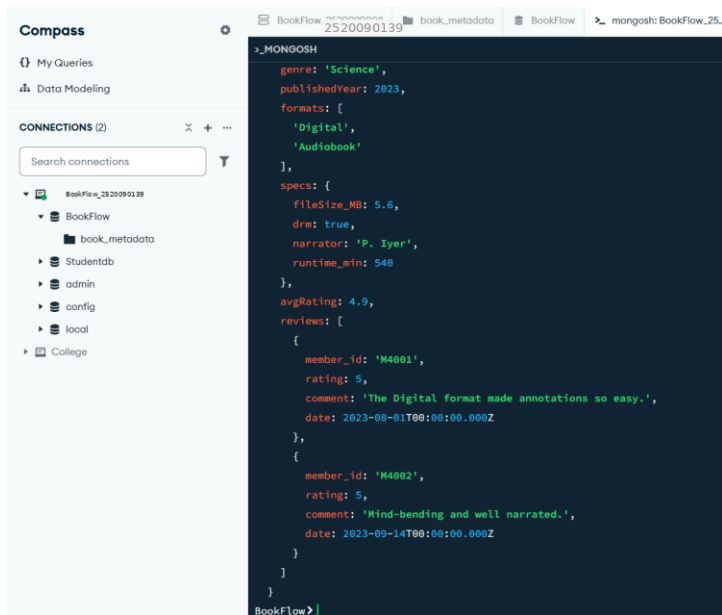
The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (2)' panel lists the 'BookFlow' database. The main panel displays a query in the 'mongosh' terminal:

```
>_MONGOSH
> db.book_metadata.find({
  avgRating: { $gt: 4 },
  formats: { $in: ["Digital"] }
})
< {
  _id: ObjectId('6a8d23960e5a8db208c7beee'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ]
}
```

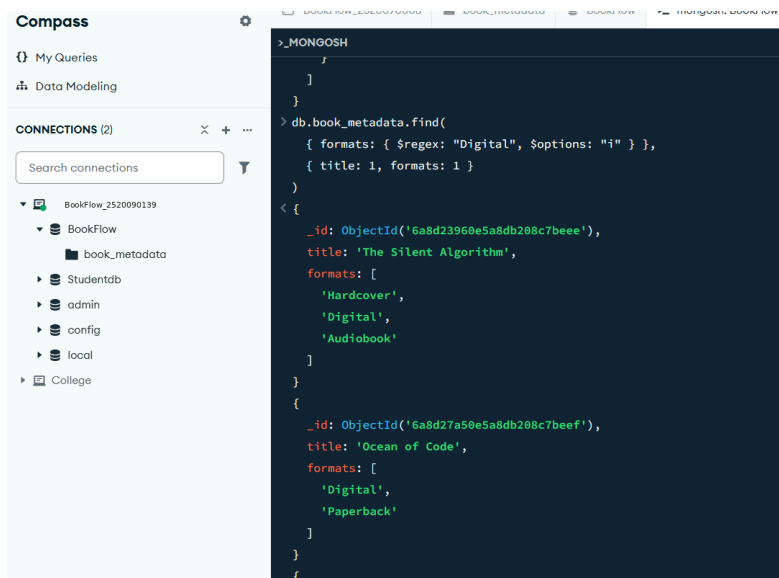


The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (2)' panel lists the 'BookFlow' database. The main panel displays a query in the 'mongosh' terminal:

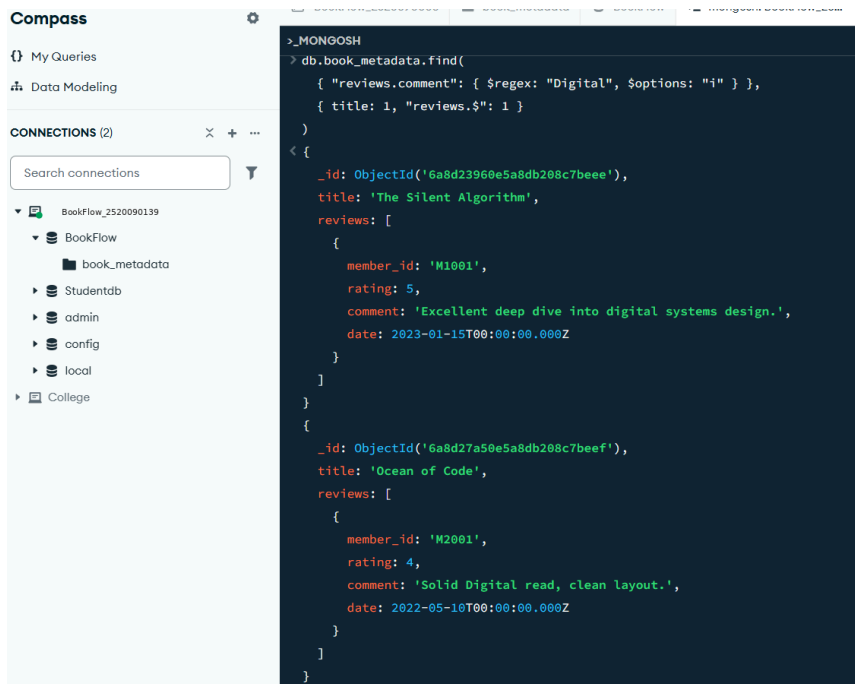
```
>_MONGOSH
> db.book_metadata.find({
  avgRating: { $gt: 4 },
  formats: { $in: ["Digital"] }
})
< {
  _id: ObjectId('6a8d27a50e5a8db208c7beef'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [
    'Digital',
    'Paperback'
  ],
  specs: {
    pages: 210,
    fileSize_MB: 2.1,
    drm: false
  },
  avgRating: 4.2,
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.'
    }
  ]
}
```



Finding Digital Books Using Regular Expression

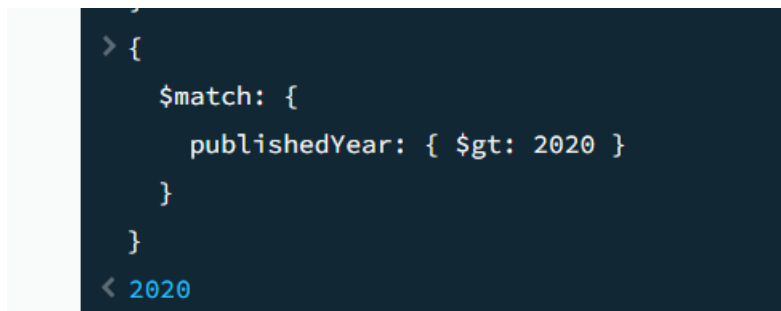


Searching Reviews Containing "Digital" Using Regex

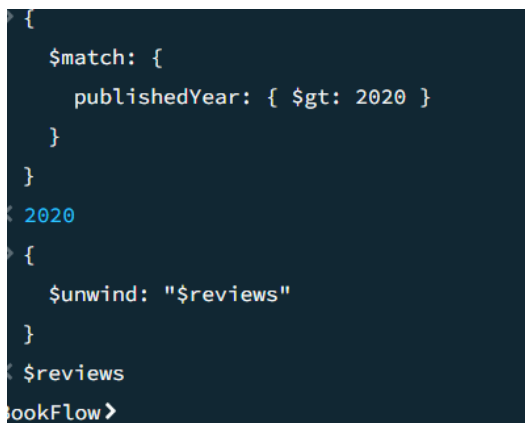


TASK – 3:

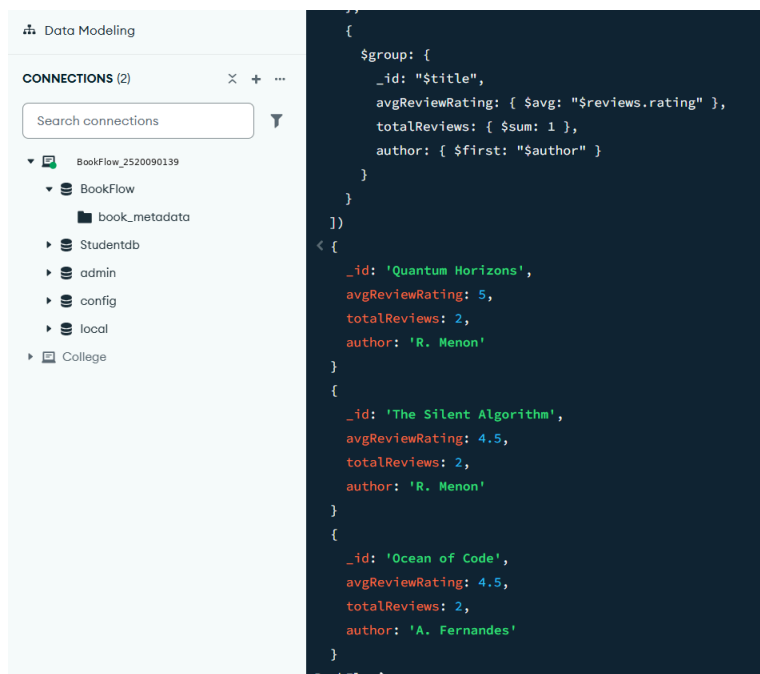
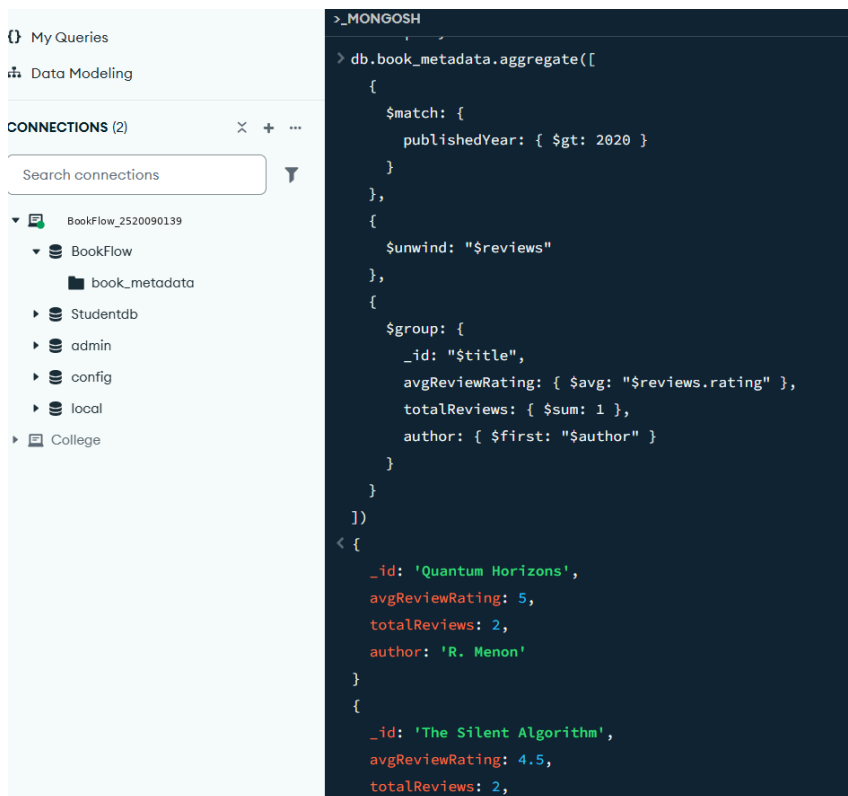
\$match — Books Published After 2020



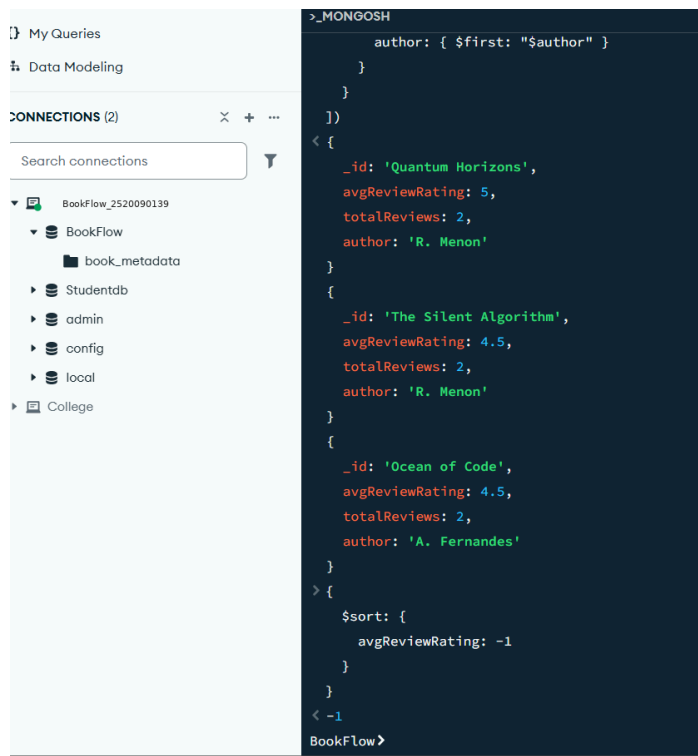
\$unwind — Expand Reviews



Grouping Books and Calculating Average Rating Using \$group



Sorting Books by Average Rating Using \$sort



Formatting the Top Rated Report Using \$project

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

BookFlow_2520090139

BookFlow

book_metadata

Studentdb

admin

config

local

College

```
>_MONGOSH
> db.book_metadata.aggregate([
  {
    $match: {
      publishedYear: { $gt: 2020 }
    }
  },
  {
    $unwind: "$reviews"
  },
  {
    $group: {
      _id: "$title",
      avgReviewRating: { $avg: "$reviews.rating" },
      totalReviews: { $sum: 1 },
      author: { $first: "$author" }
    }
  },
  {
    $sort: {
      avgReviewRating: -1
    }
  },
  {
    $project: {
      _id: 0,
      title: "$_id",
      author: 1,
      avgReviewRating: {
        $round: ["$avgReviewRating", 2]
      }
    }
  }
])
```

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

BookFlow_2520090139

BookFlow

book_metadata

Studentdb

admin

config

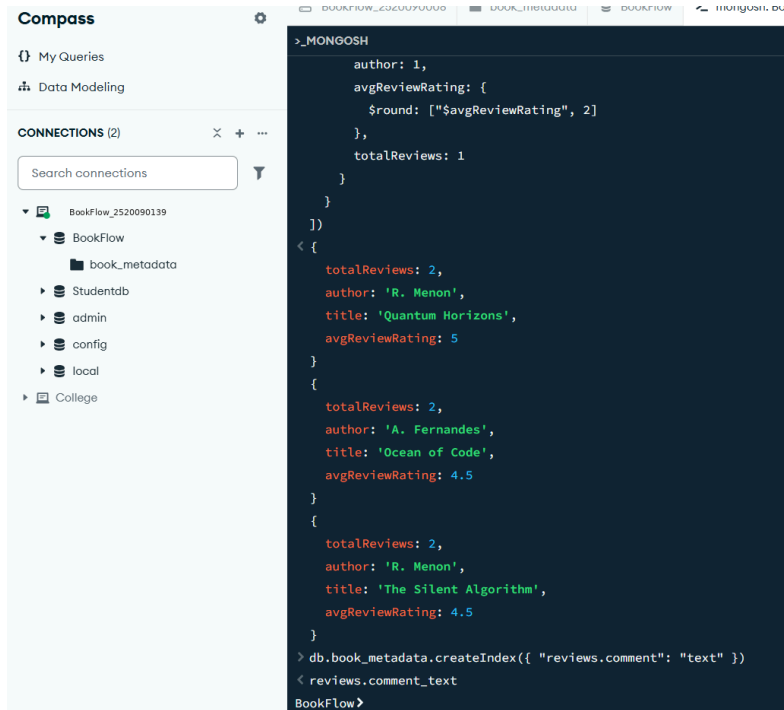
local

College

```
>_MONGOSH
  _id: 0,
  title: "$_id",
  author: 1,
  avgReviewRating: {
    $round: ["$avgReviewRating", 2]
  },
  totalReviews: 1
}
])
< {
  totalReviews: 2,
  author: 'R. Menon',
  title: 'Quantum Horizons',
  avgReviewRating: 5
}
{
  totalReviews: 2,
  author: 'A. Fernandes',
  title: 'Ocean of Code',
  avgReviewRating: 4.5
}
{
  totalReviews: 2,
  author: 'R. Menon',
  title: 'The Silent Algorithm',
  avgReviewRating: 4.5
}
}
BookFlow>
```

Task 4: Indexing for Speed

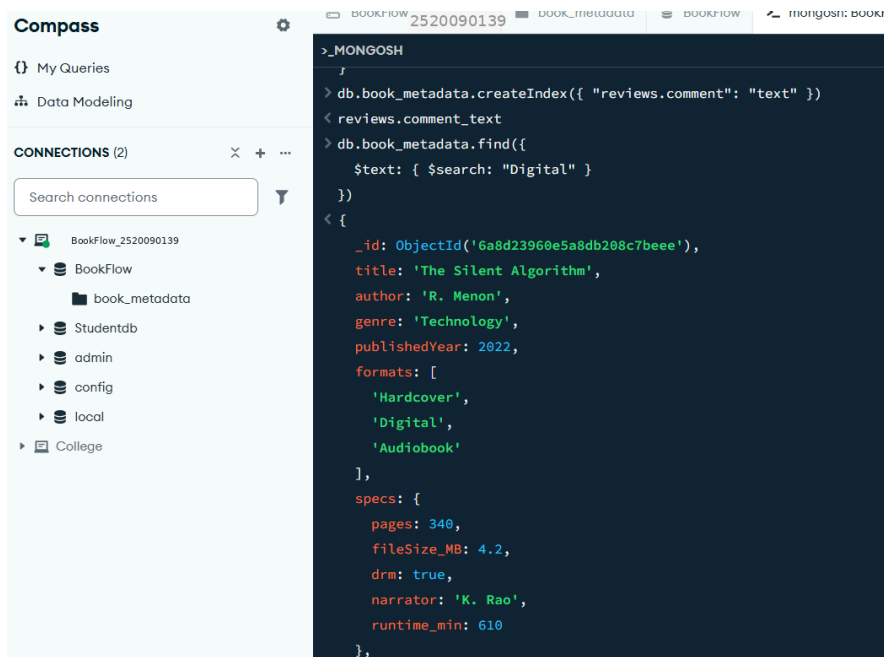
4.1 Creating a Text Index on Review Comments



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (2)' panel lists the 'BookFlow' database and the 'book_metadata' collection. The main panel displays the MongoDB shell with the following commands and results:

```
>_MONGOSH
{
  author: 1,
  avgReviewRating: {
    $round: ["$avgReviewRating", 2]
  },
  totalReviews: 1
}
})
< {
  totalReviews: 2,
  author: 'R. Menon',
  title: 'Quantum Horizons',
  avgReviewRating: 5
}
{
  totalReviews: 2,
  author: 'A. Fernandes',
  title: 'Ocean of Code',
  avgReviewRating: 4.5
}
{
  totalReviews: 2,
  author: 'R. Menon',
  title: 'The Silent Algorithm',
  avgReviewRating: 4.5
}
> db.book_metadata.createIndex({ "reviews.comment": "text" })
< reviews.comment_text
BookFlow>
```

4.2 Searching Review Comments Using the Text Index



The screenshot shows the MongoDB Compass interface. The main panel displays the MongoDB shell with the following commands and results:

```
>_MONGOSH
> db.book_metadata.createIndex({ "reviews.comment": "text" })
< reviews.comment_text
> db.book_metadata.find({
  $text: { $search: "Digital" }
})
< {
  _id: ObjectId('6a8d23960e5a8db208c7beee'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
}
```

4.3 Verifying Text Index Usage Using .explain()

Compass

{ } My Queries

Data Modeling

CONNECTIONS (2)

Search connections

- BookFlow_2520090139
 - BookFlow
 - book_metadata
 - Studentdb
 - admin
 - config
 - local
- College

```
>_MONGOSH
}
> db.book_metadata.find(
  $text: { $search: "Digital" }
).explain("executionStats")
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    },
    indexFilterSet: false,
    queryHash: '4FB85230',
    planCacheShapeHash: '4FB85230',
    planCacheKey: 'F34F3876',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'TEXT_MATCH',
      nss: 'BookFlow.book_metadata',
```

Compass

{ } My Queries

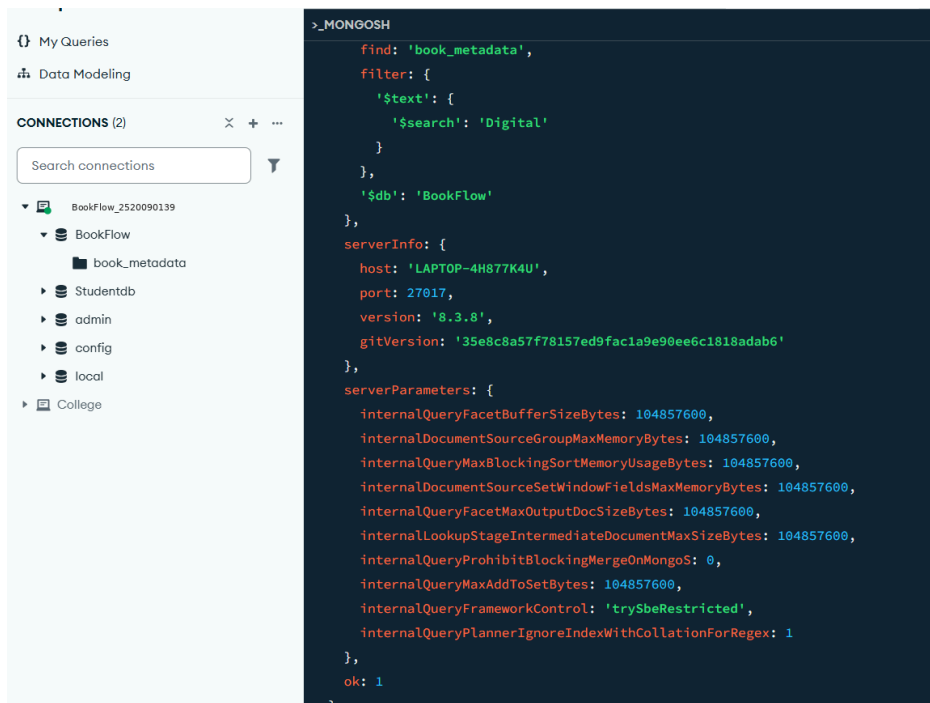
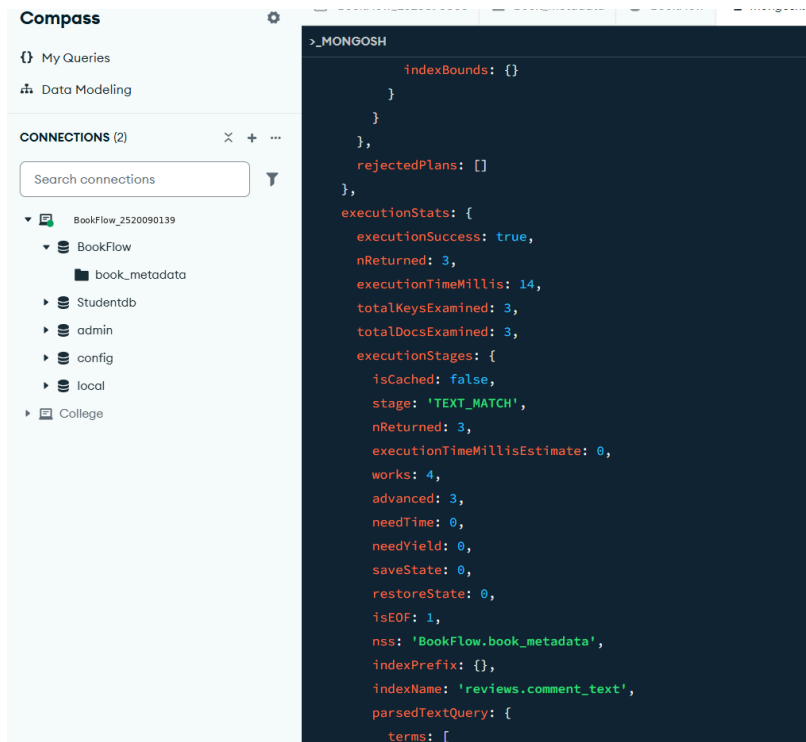
Data Modeling

CONNECTIONS (2)

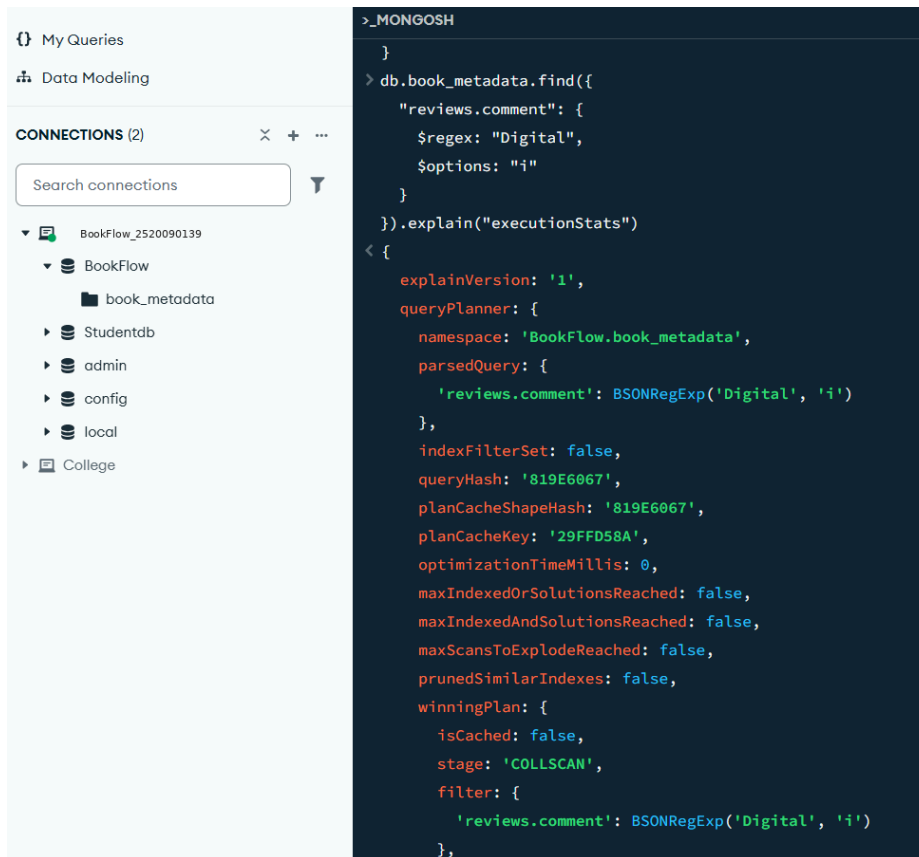
Search connections

- BookFlow_2520090139
 - BookFlow
 - book_metadata
 - Studentdb
 - admin
 - config
 - local
- College

```
>_MONGOSH
nss: 'BookFlow.book_metadata',
indexPrefix: {},
indexName: 'reviews.comment_text',
parsedTextQuery: {
  terms: [
    'digit'
  ],
  negatedTerms: [],
  phrases: [],
  negatedPhrases: []
},
textIndexVersion: 3,
inputStage: {
  stage: 'FETCH',
  nss: 'BookFlow.book_metadata',
  inputStage: {
    stage: 'IXSCAN',
    nss: 'BookFlow.book_metadata',
    keyPattern: {
      _fts: 'text',
      _ftsx: 1
    },
  },
  indexName: 'reviews.comment_text',
  isMultiKey: true,
  isUnique: false,
  isSparse: false,
  isPartial: false,
  indexVersion: 2,
  direction: 'backward',
```



4.4 Comparing Regex Search Performance Using .explain()



4.5 Creating a Multikey Index on Book Formats

```
> db.book_metadata.createIndex({ formats: 1 })
< formats_1
BookFlow>
```

4.6 Creating an Index on Average Rating

```
OK! 1
}
> db.book_metadata.createIndex({ formats: 1 })
< formats_1
> db.book_metadata.createIndex({ avgRating: -1 })
< avgRating_-1
BookFlow> |
```

4.7 Verifying Average Rating Index Using .explain()

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

BookFlow_2520090139

BookFlow

book_metadata

Studentdb

admin

config

local

College

>_MONGOSH

db.book_metadata.find({

avgRating: { \$gt: 4 }

}).explain("executionStats")

< {

explainVersion: '1',

queryPlanner: {

namespace: 'BookFlow.book_metadata',

parsedQuery: {

avgRating: {

'\$gt': 4

}

}

,

indexFilterSet: false,

queryHash: 'E21FCB34',

planCacheShapeHash: 'E21FCB34',

planCacheKey: '6A4442CC',

optimizationTimeMillis: 5,

maxIndexedOrSolutionsReached: false,

maxIndexedAndSolutionsReached: false,

maxScansToExplodeReached: false,

prunedSimilarIndexes: false,

winningPlan: {

isCached: false,

stage: 'FETCH',

nss: 'BookFlow.book_metadata',

inputStage: {

stage: 'IXSCAN',

nss: 'BookFlow.book_metadata',

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

BookFlow_2520090139

BookFlow

book_metadata

Studentdb

admin

config

local

College

>_MONGOSH

keyPattern: {

avgRating: -1

}

,

indexName: 'avgRating_-1',

isMultiKey: false,

multiKeyPaths: {

avgRating: []

}

,

isUnique: false,

isSparse: false,

isPartial: false,

indexVersion: 2,

direction: 'forward',

indexBounds: {

avgRating: [

'[inf, 4)'

]

}

}

,

rejectedPlans: []

}

,

executionStats: {

executionSuccess: true,

nReturned: 3,

executionTimeMillis: 14,

totalKeysExamined: 3,

totalDocsExamined: 3,

executionStages: {

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

BookFlow_2520090139

BookFlow

book_metadata

Studentdb

admin

config

local

College

2_MONGODB

```
find: 'book_metadata',
filter: {
  avgRating: {
    '$gt': 4
  }
},
'db': 'BookFlow'
},
serverInfo: {
  host: 'LAPTOP-4H877K4U',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
```